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SUPERSEDING
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FEDERAL SPECIFICATION

STENCIL-CUTTING MACHINE, HAND-OPERATED

This specification is by the Commissioner, Federal Supply Service, General Services Administration, for use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope. This specification covers one type of hand-operated stencil-cutting machine.

1.2 Classification.

1.2.1 Sizes. Stencil-cutting machines covered by this specification shall have the following character sizes, as specified (see 6.2): 1/4-inch, 1/2-inch, 3/4-inch, 1-inch, 1-1/4-inch, 1-1/2-inch, and 1-3/4-inch.

2. APPLICABLE STANDARDS

2.1 Standards. The following standards, of the issues in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

Federal Standards:

Fed. Std. No. 102 - Preservation, Packaging, and Packing
Fed. Test Method Std. No. 151 - Metals; Test Methods.

(Activities outside the Federal Government may obtain copies of Federal Specifications, Standards, and Handbooks as outlined under General Information in the Index of Federal Specifications, Standards, and Handbooks and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington,D.C. 20402.)

FSC 7490

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Military Standards:

MIL-STD-105-Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-130-Identification Marking of U.S. Military Property.

(Copies of Military Specifications and Standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Pre-production sample. When specified (see 6.2), before production is commenced, a sample of the finished item shall be made available to the contracting officer or his authorized representative for approval in accordance with 4.2. The approval of the pre-production sample authorizes the commencement of production but does not relieve the supplier of responsibility for compliance with all applicable provisions of this specification. The pre-production sample shall be manufactured in the same facilities to be used for the manufacture of the production items.

3.2 Standard product. The stencil-cutting machine shall be a standard current product of the manufacturer except for any deviations specified herein.

3.3 Materials. Material used shall be of the quality used in acceptable commercial practice.

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3.4 Design. The stencil-cutting machine shall be of the portable, circular type with a base designed for attaching to a table or bench. The stencil-cutting machine shall be designed to cut stencils of any desired length with a capacity of not less than four lines. The stencil-cutting machine shall produce clear, legible, equally spaced characters when tested as specified in 4.4.1. All stencil-cutting machines, except sizes 1-1/4-, 1-1/2-, and 1-3/4-inch characters shall be completely housed and covered. Sizes 1-1/4-, 1-1/2-, and 1-3/4-inch character stencil-cutting machines shall be semi-housed.

3.5 Character dial. The character dial shall have clearly legible characters defined in a permanent manner over the entire range. Each stencil-cutting machine shall be equipped with a pointer indicator to clearly identify the character selected on the character dial.

3.6 Characters. Each stencil-cutting machine shall be designed to cut the 40 characters as follows:

(a) 26 letters of the alphabet.
(b) Figures "2" to "9".
(c) Comma.
(d) Period.
(e) Abbreviation for "and".
(f) Apostrophe.
(g) Dash.
(h) Slant used in fractions.

The styles of the letters and figures may be either Roman or Gothic, as specified (see 6.2). A plus or minus tolerance of 1/16 inch will be allowed in the size of the letters and figures. However, each character on all stencil-cutting machines furnished under an individual contract or order shall be uniform in size and style.

3.7 Punches and dies. Punches and dies shall be fitted and mated so that sharp edges will not blunt with constant use under ordinary service conditions. The punches shall be in perfect alignment with the die and work with a sheer action to assure clean-cut edges of the character. The punch and dies shall be not less than 5/64 inch thick. The punch and dies shall be not less than 5/64 inch thick. The punch and dies shall be easily replaceable with the use of ordinary tools.

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3.7.1 Punches. The punches shall be fabricated of steel or non-ferrous alloy having a hardness from Rockwell B70 to B90. There shall be no evidence of hardness requirement failure when tested as specified in 4.4.2.

3.7.2 Dies. The dies shall be fabricated of steel, surface hardened to Rockwell Superficial 15N-87 minimum. There shall be no evidence of hardness requirement failure when tested as specified in 4.4.2.

3.8 Handle. The handle shall have sufficient leverage to operate the stencil-cutting machine freely with a wrist action when tested in accordance with 4.4.1.

3.9 Attachment. When specified (see 6.2), the stencil-cutting machine shall be provided with an attachment for cutting circular stencils. Characters shall be as specified in 3.6.

3.10 Spacing. The stencil-cutting machine shall be equipped with a mechanism to automatically and accurately space each character, and to provide a space on the stencil by a partial movement of the cutting handle when tested in accordance with 4.4.1. The platform that holds the stencil board in place shall be provided with legible, graduated guidelines, permanently marked, or some other means shall be provided for quick alignment and setting of the stencil.

3.11 Lubrication. Provisions shall be made so that bearing surfaces requiring lubrication are accessible and readily lubricated.

3.12 Supplies. Unless otherwise specified (see 6.2), the following supplies shall be furnished and packed with each stencil-cutting machine:

(a) One brush or roller and pad, or one brush or roller with replacement cartridge which is intended to be used as the handle.
(b) A minimum of one pint of black stencil ink. When furnished in the cartridge form as stated in (a) above, a sufficient number of cartridges shall be furnished to provide a total volume of not less than one pint of ink.
(c) One cleaning hook.
(d) Ten pounds of good commercial quality stencil board which shall be of a size for use with stencil-cutting machine.
(e) Complete instructions for its operation and care.
(f) Parts list.

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3.13 Component parts and assemblies. The type and size stencil-cutting machines, hand operated, furnished to the Government under this specification on any one contract will be identical and each specific component part and assembly thereof will be identical.

3.14 Marking for identification.

3.14.1 Civil Agencies. Each stencil-cutting machine shall be marked in a plain and permanent manner with the name of the manufacturer or trademark of such known character that the source of manufacture can be readily identified.

3.14.2 Military Agencies. Each stencil-cutting machine shall be marked in accordance with MIL-STD-130.

3.15 Finish. Unless otherwise specified in the contract or order, all exposed parts shall be given a durable coat of black or other dark-color paint or enamel, and provisions shall be made to protect all uncoated surfaces from rust and corrosion (see 6.2).

3.16 Workmanship. The finished commodity shall conform to the quality of product established by this specification.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specifications where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. (See 6.2)

4.2 Pre-production sample inspection. When a pre-production sample is required, the pre-production sample shall be examined in accordance with the defects in 4.3.3.1 and for defects in dimensions. The pre-production sample shall be tested in accordance with 4.4.1.

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4.3 Sampling for inspection and acceptance. Sampling for inspection and acceptance shall be performed in accordance with the provisions set forth in MIL-STD-105, except where otherwise indicated herein.

4.3.1 Inspection of materials and components. In accordance with 4.1, the supplier is responsible for insuring that materials and components used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein, or, if none, in accordance with this specification. In the event of conflict, this specification shall govern.

4.3.1.1 Component testing. Testing shall be performed on components listed in table I for characteristics shown therein. A lot shall consist of all punches and dies submitted for testing at one time.

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TABLE 1. Testing of components

Component	Characteristic	Requirement paragraph	Test method paragraph	Requirements applicable to--- Individual unit	Requirements applicable to--- Lot average	Number determinations per sample	Results reported as---	Sample unit	Inspection level	Acceptance number
Punches	Hardness Rockwell B70 to B90	3.7.1	4.4.2	X	---	3	Separate Readings	1	S-4	0
Dies	Hardness Rockwell Superficial 15N-87, minimum	3.7.2	4.4.2	X	---	3	Separate Readings	1	S-4	0

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4.3.2 In-process examination. Examination shall be made during manufacture to assure that punches and dies are properly fitted and mated and that punches are in alignment in accordance with 3.7. Any failure to comply shall be cause for rejection of the lot in process.

4.3.3 End item inspection. Examination shall be made of the completely assembled stencil-cutting machine. The inspection lot shall consist of all stencil-cutting machines of one size character offered for inspection at one time. The sample unit for these examinations shall be one stencil-cutting machine.

4.3.3.1 Visual examination. Examination of the stencil-cutting machine shall be in accordance with the classification of defects in table II. The inspection level shall be level II with an acceptable quality level (AQL) of 1.5 for major defects and 4.0 for total defects, expressed in terms of defects per one hundred units.

Table II. Classification of end item defects

Examine---	Defects	Classification	
		Major	Minor
Finish.	Peeling or blistering		X
	Not finished as specified		X
	Sags, runs, orange peel, wrinkles		X
	Dirt, grit, or foreign matter imbedded in finish .		X
	Finish tacky (not completely dry)		X
	Not dark color		X
Design.	Evidence of rust or corrosion	X	
	Characteristic not in accordance with the specified requirements	X	
Components.	Fractured, split, malformed, or otherwise damaged.	X	

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Construction and workmanship.	Sharp burr or sliver on surfaces to be handled during operation.	X	
	Functioning component that is inoperative or will not function as intended	X	
Character dial.	Not clear legible characters	X	
	Pointer does not clearly identify the the character selected.	X	
Lubrication.	Bearing surfaces requiring lubrication not accessible or easily lubricated	X	
Spacing.	Not as specified. ..	X	
Marking.	Missing, illegible, or otherwise incorrect.	X X	

4.3.3.2 Dimensional examination. Inspection shall be made for compliance with dimensions specified in applicable paragraphs. Any deviation therefrom shall constitute a defect. The inspection level shall be level II with an AQL of 1.5 expressed in terms of defects per one hundred units.

4.3.3.3 Testing of the end item. Testing shall be performed on each sample unit for characteristics specified in 4.4.1. The inspection level shall be S-4 with an AQL of 1.5 expressed in terms of defects per one hundred units.

4.3.4 Inspection of preparation for delivery requirements. An inspection shall be made to determine that packaging, packing and marking requirements of section 5 of this specification are complied with. Defects shall be scored in accordance with table III. For examination of contents the sample unit shall be one shipping container fully prepared for delivery selected just prior to the closing operations. Defects of closure listed shall be examined on shipping containers fully prepared for delivery. The lot size shall be the number of shipping containers in the end item inspection lot. The inspection level shall be S-2 and the AQL shall be 4.0 defects per one hundred units.

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TABLE III. Inspection of preparation for delivery defects

Examine--	Defects
Markings (exterior and interior).	Omitted; incorrect; illegible; or improper size, location, sequence, or method of application.
Materials.	Any component missing, or damaged.
Workmanship.	Inadequate application of components such as incomplete closure of container flaps, loose strapping, inadequate stapling. Bulging or distortion of container.

4.4 Tests.

4.4.1 Performance test. The stencil-cutting machines shall be tested with stencil board under operating conditions using all characters and spacing to determine compliance with 3.4, 3.8, and 3.10.

4.4.2 Hardness test. The Rockwell hardness for the punches and dies shall be tested in accordance with method 243 of Fed. Test Method Std. No. 151 to determine compliance with 3.7.1 and 3.7.2.

5. PREPARATION FOR DELIVERY

(For civil agency procurement the definitions and application of levels of packaging and packing shall be in accordance with Fed. Std. No. 102.)

5.1 Preservation and packaging. Unless otherwise specified in the contract or order the stencil-cutting machine shall be packaged in accordance with ASTM D3951 Commercial Packaging, Standard Practice for.

6. NOTES

6.1 Intended use. This specification covers stencil-cutting machines intended to be used in cutting stencil board or cardboard stencils.

6.2 Ordering data. Purchasers should exercise any desired options offered herein (see 3.15, 4.1) and procurement documents should specify the following:

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(a) Title, number, and date of this specification.
(b) Character size required (see 1.2.1).
(c) When pre-production sample is required (see 3.1)
(d) Specify if Gothic or Roman characters are required (see 3.6)
(e) Circular stencil-cutting attachment if required (see 3.9)
(f) When supplies are not required (see 3.12).
(g) Finish, if other than specified (see 3.14).
(h) Level of packaging and packing required (see 5.1).

6.3 Repair parts. The contracting officer should determine whether repair parts are required as part of the contract or order. When repair parts are required, the contract or order should specify the following:

(a) Manufacturer's name or trademark of the stencil-cutting machine for which the repair parts are required.
(b) Proper name of the part required.
(c) Quantity of each item required.

6.3.1 When repair parts are not required with the contract or order, the contracting officer should require bidders to furnish assurance that parts are available for future procurement.

6.4 Transportation description. Transportation descriptions and minimum weights applicable to this commodity are:

Rail:

Machines, stencil cutting, with stand knocked down, or without stands. Carload minimum weight 24,000 pounds, subject to Rule 34, Uniform Freight Classification.

Motor:

Machines, stencil cutting, with stands knocked down, or without stands. Truckload minimum weight 24,000 pounds, subject to Rule 115, National Motor Freight Classification.

6.5 Federal specifications do not include all types, classes, grades, sizes, etc., of the commodities indicated by the titles of the specifications, or which are commercially available, but are intended to cover the types, etc., which are suitable for Federal Government requirements.

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MILITARY INTERESTS:

NONE: DoD has determined that no military activity has an official interest in this Federal specification.

CIVIL AGENCY COORDINATING ACTIVITIES:

GSA-FSS

PREPARING ACTIVITY:

GSA-FSS-FCGC

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